

The impact of business process management and applicant tracking systems on recruiting process performance: an empirical study

Sven Laumer · Christian Maier · Andreas Eckhardt

Published online: 13 December 2014
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Abstract This research focuses on the effects of different business process management components in combination with information technology on recruiting process performance. The results of a study of Germany's largest 1,000 business enterprises (response rate 13.1 %) reveal that business process analysis, business process improvement and the usage of applicant tracking systems reduce recruiting process costs. Specifically, the cycle time of the recruiting process can be shortened significantly through business process controlling and process analysis, and by using an applicant tracking system that supports the design and evaluation of key performance indicators. Business process standardization combined with applicant tracking systems and business process documentation as well these systems used together with business process controlling have a significant positive impact on stakeholder satisfaction with the recruiting process. The general quality of the process can be improved through business process controlling as well as through a combination of applicant tracking systems and business process controlling. Our results reveal that several components of the business process management in conjunction with a supporting applicant tracking system have differing impacts on recruiting process performance. This paper discusses these diverse effects of

S. Laumer (✉) · C. Maier

Lehrstuhl für Wirtschaftsinformatik, insb. Informationssysteme in Dienstleistungsbereichen,
Centre of Human Resources Information Systems, Otto-Friedrich Universität Bamberg,
An der Weberei 5, 96047 Bamberg, Germany
e-mail: sven.laumer@uni-bamberg.de

C. Maier

e-mail: christian.maier@uni-bamberg.de

A. Eckhardt

Institut für Wirtschaftsinformatik, Centre of Human Resources Information Systems,
Goethe-Universität Frankfurt am Main, Grüneburgplatz 1, 60323 Frankfurt, Germany
e-mail: eckhardt@wiwi.uni-frankfurt.de

business process management on process performance and draws implications for information systems success research.

Keywords Business process management · Recruiting · Process performance · Applicant tracking system · Human resources information systems · Information systems success

JEL Classification M12 · M15 · O30 · O31

1 Introduction

Business process management (BPM) enables companies to leverage the effectiveness and efficiency of business processes and to gain competitive advantages (Hung 2006). BPM has been the subject of intense discussion in information system (IS) research and by practitioners since the late 1980s and has been applied primarily in the industry context focusing on customer-facing or operational processes (Lee 1998). The traditional focus of BPM is on primary business processes, whereas BPM approaches focusing on secondary service processes are seldom the focus of academic studies and are often neglected by managers in companies (Ulbrich 2006; Kohlbacher 2010). For example, for the secondary support process of human resources management (HRM), “no one seems to have developed a business process model of the HRM business processes” (Cakar et al. 2003, p. 191), although “treating the HRM process as a support process providing critical support to the key valued-add processes” (Cakar et al. 2003, p. 191).

Nonetheless, even HR departments, which have been the organizational business units most averse towards information technology (IT) for decades, have been increasingly implementing and using IS (Strohmeier 2009) especially in staff recruitment for their daily business for some years now (Lee 2007; Eckhardt et al. 2012, 2014; Holm 2012; Wiblen et al. 2012). Moreover, strategic approaches in managing these business processes have been implemented by companies to achieve better process performance in terms of time, cost, quality, and satisfaction (Münstermann et al. 2010a, b; Eckhardt et al. 2014). That is, recruiting departments have also started to focus on BPM and IT support to improve the performance of the recruiting process, just as operations departments have been doing with organizational core processes during the last decades. However, empirical studies investigating the effects of implementing BPM and applicant tracking systems (ATS) for the recruiting process are still rare.

Kohlbacher (2010) points out that it would be interesting to investigate which BPM components have an effect on which outcome variables. Prior research has not always explicitly differentiated between the deployment of various BPM components and their effects on process performance. Such a differentiation would help companies to implement the BPM components that most impact process performance. Kohlbacher (2010) criticizes prior case study-based research of the effects of BPM, pointing out that “case study reports do not always deliver exact information

about the deployment of the individual [BPM] dimensions” (p. 14) and identifying “a clear lack of quantitative studies exploring the effects of [BPM]” (p. 14). This research aims to partially fill this gap by studying the various effects of different BPM components on recruiting process performance based on a representative empirical study of the 1,000 largest business enterprises (herein referred to as “companies” in the general sense of the word) in Germany to provide more general empirical results of the effects of BPM on process performance, as demanded by Kohlbacher (2010). Hence, the research question of this research article focusing on BPM in the recruiting context is:

What are the effects of different BPM components and IT on recruiting process performance?

In the following section, we introduce the concept of BPM and the recruiting process in preparation for developing a research model of the impact of BPM and IT on recruiting process performance in the section afterwards. The subsequent section presents the methodology and the results of our survey and the results of our research approach will be discussed in the concluding section of the paper.

2 Concepts

In this section, we will introduce BPM concepts and provide essential information about the recruiting process.

2.1 Business process management

BPM focuses on business processes which are “a collection of interrelated events, activities and decision points that involve a number of actors and objects, and that collectively lead to an outcome that is of value to at least one customer”. (Dumas et al. 2013, p. 3). BPM “can generally be viewed as a collection of process improvement efforts that differ in mission, scope, and approach” (Ho et al. 2009, p. 2). It “relies on systems and documented procedures to ensure discipline, consistency, and repeatability of quality performance” (Zairi 1997, p. 65). There are many disparate views regarding BPM (Isaksson 2006; Palmberg 2009). Especially with regard to the system level, the definitions used vary widely (Biazzo and Bernardi 2003). In addition, scholars and practitioners make divergent recommendations on BPM approaches and tools and do not give clear and consistent guidance on how to deploy BPM (Palmberg 2009).

In general, BPM approaches start by identifying a company’s processes to develop a process architecture. Based on this inventory of the company’s processes, the core processes for improvement activities are identified (Dumas et al. 2013). The core process owners and the BPM team document and analyze the processes, identify performance problems, select an improvement strategy, and implement changes to the process (Elzinga et al. 1995; Forsberg et al. 1999; Biazzo and Bernardi 2003; Hung 2006; Houy et al. 2010). Thus, the processes are mapped to derive as status quo process model (Dumas et al. 2013), which is analyzed for weaknesses. Based on the weaknesses identified, the process is redesigned and a

target process model is developed (Dumas et al. 2013). Finally, the redesigned process is implemented following an executable process model (Dumas et al. 2013). BPM also includes a continuous monitoring of process performance, assessment of results, and continuous implementing improvements (Zairi 1997; Isaksson 2006; Ulbrich 2006; Ugan 2006; Palmberg 2009; Münstermann et al. 2010a, b; Weske 2012). These performance insights can be used to restart the described BPM lifecycle (Dumas et al. 2013). Hence, BPM is a set of tools and techniques for improving processes and a method for integrating the entire company with a focus on its processes (Harmon 2007; Rummmler and Ramias 2010; Becker et al. 2011; Röglinger and Kamprath 2012).

Consequently, BPM can be either defined as “a structured systematic approach to analyze and continually improve the process” (Palmberg 2009, p. 204), which Houy et al. (2011) refer to as “BPM in the large”, or as “a holistic manner to manage all aspects of the business and a valuable perspective in determining organizational effectiveness” (Palmberg 2009, p. 204), which Houy et al. (2011) call “BPM in the small”. According to this distinction, “BPM in the small” concentrates on single processes or scenarios, comprising a few isolated processes, while “BPM in the large” focuses on handling complexity and dynamics throughout the entire company or the whole BPM life cycle including collaborative modeling, configurable process models, and process mining. As we are investigating the effects of implementing BPM components in only one particular process, the recruiting process performance, we have adopted the definition of “BPM in the small” (Palmberg 2009; Houy et al. 2011) and the components used to implement “BPM in the small”. We thus define BPM in keeping with Dumas et al. (2013) as a body of methods, techniques and tools to discover, analyze, redesign, execute and monitor business processes.

2.2 Recruiting process

The recruiting process is widely considered one of the most time- and cost-consuming human resources processes (Münstermann et al. 2010b). Companies increasingly recognize the importance of and need for better coordination and management of the numerous tasks in staff recruitment (Wiblen et al. 2010; Holm 2012). There is a particular need for process management in the recruiting process because this process is anything but straightforward and has grown dramatically more complex due to external developments (e.g., demographic change, global talent shortage; Worth 2011), strategic renewals (e.g., a holistic talent management approach; Ashton and Morton 2005), as well as technological innovation (Lee 2007; Eckhardt and Laumer 2009; Laumer et al. 2009a, b, 2010a, b) over the last years. Moreover, both actors in the recruiting process—recruiter and applicant—are highly complex individuals (Laumer et al. 2009a, b, 2010a, 2012; Maier et al. 2013) who act according to numerous different behavioral patterns that are hard to control for (Laumer et al. 2010a; Holm 2012). Furthermore, applicants contact the hiring company via several different media and apply for jobs using different forms of application (Wiblen et al. 2012; Eckhardt et al. 2014). To respond to the needs of these different groups of applicants, most companies need to develop significantly

differing recruiting process forms to meet their overall staff recruitment needs. A higher number of variants leads to higher overall process complexity and the strong need of BPM for this secondary business process (Eckhardt et al. 2014).

Even within companies, recruiters do not act according to fixed standard patterns, but rather often take an individual recruiting process approach (Holm 2012). Recommendations and prior experiences of their workplace referents as well as their individual attitude are all significant determinants for their particular behavior (Eckhardt et al. 2009; Wiblen et al. 2010). The behavioral patterns vary widely, ranging from completely standardized behavior supported by an underlying applicant tracking system (ATS) to completely unstructured and frequently changing behavior without a straightforward process (Lee 2007; Holm 2012). Hence, the recruiting process is highly unstandardized even in many large companies, which makes it particularly suitable for our research purpose.

In general, the recruiting process contains six steps (Lee 2007; Holm 2012). In the first step, “employer branding” a company tries to develop a particular image through marketing activities (Tumasjan et al. 2011). The company’s objective is to project an image of itself that matches the applicant’s imagined ideal employer. In the second step, “candidate attraction”, companies try to fill their vacancies within a short time frame (Wiblen et al. 2012). This typically includes posting job ads across on- and offline channels (Baum and Kabst 2012). In the “applicant management” step, companies manage incoming applications and communications with applicants. After the following “pre-selection” and “selection” steps, the recruiting process ends with the final “hire” (Eckhardt and Laumer 2009). The overall recruiting process is illustrated by Fig. 1.

Based on the definitions of BPM discussed above and the description of the recruiting process as an example of a complex secondary business process, the following section introduces our research model for investigating the impact of BPM and applicant tracking systems on recruiting process performance.

3 Research model

In this section, after defining the dependent variable recruiting process performance, we hypothesize the effects of BPM and IT on recruiting process performance, develop the research model, and discuss each hypothesis of the entire model in turn.

3.1 Recruiting process performance

In general, BPM has a variety of potential benefits. For example, Hammer (2007, p. 111) observes that “in virtually every industry, companies of all sizes have achieved extraordinary improvements in cost, quality, speed, profitability, and other



Fig. 1 Recruiting process (Lee 2007; Holm 2012)

key areas by focusing on, measuring, and redesigning their customer-facing and internal processes". Karimi et al. (2007) suggest defining business process performance as process efficiency and effectiveness. In this context, Hirzel (2008) names lower costs, shorter cycle time, greater customer satisfaction, and higher quality as the main benefits companies can realize by making processes more efficient and effective through BPM. Moreover, Schmelzer and Sesselmann (2003) and others [see (Kohlbacher 2010) for an overview] report that BPM can yield positive effects such as faster work speed, higher stakeholder satisfaction, higher quality, lower costs, and better financial performance.

We consider *Recruiting Process Performance* as the dependent variable in our research model, defined along the well-established dimensions of process cycle time, process costs, process quality, and process stakeholder satisfaction (see Table 1).

The most common process dimension in staff recruitment is *process time*, which is defined as the period of time between identifying and filling an open position (Hirzel 2008). Process time is also widely known as time-to-hire. The cycle times of single process steps include the time span between the identifying an open position and posting a job ad, the time span between posting a job ad and receiving the first applications, and the time spent on the internal applicant tracking process (Münstermann et al. 2010a; Eckhardt and Laumer 2009).

The second most common performance dimension in recruiting is *process cost*, which is defined as the aggregate cost incurred in hiring a new employee. The total process cost in recruiting is sometimes referred to as cost-per-hire. Apart from total process cost, several other costs for single process steps are traceable. For example, companies frequently trace the cost of and calculate the cost advantage of online job ads compared to offline job ads (e.g., in newspapers). Companies also estimate and trace the cost structures for the internal applicant tracking process as well as the single cost for every tracked application (Münstermann et al. 2010a; Eckhardt et al. 2012).

The third performance dimension in recruiting, *process quality* typically refers both to the quality of the applicant as well as to how satisfied stakeholders are with the newly hired candidate and with the entire recruiting process. To more clearly distinguish these dimensions, we therefore distinguish between process quality and

Table 1 Recruiting process performance

Dimension	Definition
Process time	Reduced cycle time of the recruiting process, i.e. time-to-hire, the time to published a job ad, and the time to interview is reduced and a hiring decision can be made faster than in the past
Process cost	Reduced cost of the recruiting process, i.e. lower cost for employer branding, applicant attraction, candidate management, and selection. For example, the cost per application, the cost per hire, and the cost of managing applications can be reduced
Process quality	Increased quality of the recruiting process in terms of overall quality of hired candidates, increased quality of candidates, increased quality of applications and information available to make a decision
Stakeholder satisfaction	Increased satisfaction of recruiting process stakeholders like hiring managers, HR managers, and candidates

stakeholder satisfaction. Process quality can be understood as the number of desired candidates successfully hired as well as the ratio of the quality of hired candidates from 1 year to the next. A further indicator of process quality is an increase in the quality of applicant data (e.g. the information provided in an application) (Münstermann et al. 2010a; Eckhardt et al. 2012).

The final quality-related performance dimension is *stakeholder satisfaction* with the overall recruiting process. Stakeholders include company employees, applicants, operating departments, HR executives and top management (Hirzel 2008).

Based on these dimensions of recruiting process performance, the following section discusses the impact of BPM and IT in the recruiting process in terms of their potential to improve process time, reduced costs, and increased quality and satisfaction.

3.2 Business process management as driver of recruiting process performance

The previous section introduced the distinct concepts of BPM and the recruiting process and discussed the performance dimensions of the recruiting process. The following section combines these different approaches and proposes several hypotheses for the impact of BPM on recruiting process performance.

As discussed above, we adopt the definition of BPM as a “BPM in the small” concept, which allows us to propose a separate hypothesis of the impact on recruiting process performance for each BPM component. In order to identify the components of BPM that may impact recruiting process performance, it is first important to understand and define the process, its key activities, and its architecture (Houy et al. 2011). These activities correspond with the process identification and discovery phase of the BPM lifecycle (Dumas et al. 2013). Identifying performance measurements and targets used in analyzing the process to identify and implement process improvement measures is another key activity of BPM on the operational level (Palmberg 2009). This activity corresponds with the process analysis step of the BPM lifecycle (Dumas et al. 2013). In order to redesign, further standardize, better govern or otherwise improve a business process, identified improvements have to be implemented, which corresponds with the process redesign and implementation steps of the BPM lifecycle. Finally, the executable business process has to be monitored and controlled to derive conformance and performance insights and continuously improve BPM (Dumas et al. 2013).

Our conceptualization model of BPM is based on these definitions. Its components are summarized in Table 2 and their individual effects on recruiting process performance are discussed in detail below.

3.2.1 The impact of business process documentation on process performance

The first BPM dimension “business process documentation” reflects the characteristic of BPM that all “major activities have to be properly mapped and documented” (Zairi 1997, p. 65). This dimension is defined as “all documents describing the business process including work flows, rules, data, as well as task and decision responsibilities” (Beimborn and Joachim 2011, p. 341). Documentation is

Table 2 Business process management components

BPM dimension	Description
BP documentation	Business process documentation shows the relations between the activities, personnel, information, and the objectives in a distinct workflow
BP analysis	Business process controlling identifies performance measurements and targets for controlling on the operational level which allow process improvements to be identified and implemented
BP standardization	Business process standardization includes a goal-directed homogenization (e.g., reduction of process variants) and realization of economies of scale for process bundling
BP governance	Business process governance includes the ownership of a distinct process to determine responsibility and leadership
BP controlling	Business process controlling involves diagnosing and analyzing business process execution
BP improvement	Business process improvement targets the continuous enhancement of a company's distinct business process including its disciplined design and careful execution

an important driver of recruiting process performance because it enables a company to identify inefficient business processes by designing and analyzing business process documents that serve as the foundation of optimization, change, and standardization activities (Ungan 2006; Beimborn and Joachim 2011). Among others, media disruptions can be identified and eliminated more easily while documenting business processes, which reduces process cycle time. The process discovery activities involved in business process documentation or modeling are an important step in the BPM lifecycle and contribute to efficient and effective business processes (Sharp and McDermott 2009; Dumas et al. 2013).

In the recruiting context, documentation is the basis for successfully designing and performing the recruiting process. For example, documenting and defining process steps is the basis for identifying successful recruiting channels, implementing different channels for applicants to apply for a vacancy, defining the tasks and the workflows of applicant management, and selecting candidates (Lee 2007). Documenting the recruiting process formalizes the explicit and implicit knowledge of process stakeholders (Ungan 2006; Dumas et al. 2013), making implicit knowledge available to all stakeholders and improving process performance (Dumas et al. 2013). When all HR managers have access to documented experiences of different recruiting measures in all steps of the recruiting process, they are better positioned to choose strategies and use different recruiting measures. For example, companies are able to reduce process times when there is documentation on which channels effectively attract potential and qualified candidates. When companies focus solely on documented successful channels, the cost of using inefficient channels can be reduced (Wiblen et al. 2012). Moreover, the process of documenting the recruiting process enables the different stakeholders to discuss the different process steps and to identify inefficiencies (Ungan 2006). When the workflows of different tasks are well established and the different working routines are documented, the coordination of different participants can be improved such that

the process will be performed at a faster pace (Holm 2012). Furthermore, when companies document the required skills and capabilities of vacancies, the application information quality can be improved (Wiblen et al. 2012). Based on the documentation, a structure approach can be followed to collect the information necessary to make a hiring decision. Hence, for each process step, documenting the tasks and activities enables the process to be improved. Thorough documentation and the resulting more focused recruiting process enable companies to achieve their recruiting targets more quickly and cost-efficiently. Hence, we hypothesize that:

Hypothesis 1a: *The higher the degree of business process documentation, the better the performance of the recruiting process.*

3.2.2 The impact of business process analysis on recruiting process performance

As “BPM relies on measurement activity to assess the performance of each individual process” (van der Aalst et al. 2003, p. 1023) *business process analysis* is also considered an important BPM component. It covers those “aspects neglected by traditional workflow products (e.g., diagnosis, simulation, etc.)” (van der Aalst et al. 2003, p. 1023). It involves analyzing existing business processes in order to identify problems, inefficiencies, bottlenecks, or media disruptions (Dumas et al. 2013). The goal of business process analysis is to identify, document and whenever possible quantify inefficiencies associated with the process. The output of this phase is a structured collection of issues which have to be addressed in process redesign activities (Dumas et al. 2013).

In the recruiting process, business process analysis increases process performance by identifying inefficiencies in the company. Inefficiencies causing idle time or delays (e.g., bottle-necks in terms of departments receiving a lot of applications in a short period of time) increase the overall cycle time of the recruiting process and ineffective resource allocation (e.g., low-yield personnel marketing channels) increase the overall cost. Hence, analyzing the recruiting process to identify bottlenecks or cost drivers will improve the performance of the recruiting process and enable the company to attract better candidates through more focused efforts and higher quality processes. Furthermore, the better the recruiting process is analyzed quantitatively and qualitatively, the more transparent it will be for external and internal stakeholder groups, which in turn increases their satisfaction. Thus, we hypothesize that:

Hypothesis 1b: *The higher the degree of business process analysis, the better the performance of the recruiting process.*

3.2.3 The impact of business process standardization on recruiting process performance

Business process standardization is one aspect of process redesign in the BPM lifecycle (Dumas et al. 2013) which defined by Shaw et al. (2007) as summarized by Münstermann et al. (2010a, p. 31) “a means to change business processes from where they are to a standard business process” The objective of business process

standardization is “to make process activities transparent and achieve uniformity of process activities across the value chain and across firm boundaries” (Wüllenweber et al. 2008, p. 213). For this paper, we follow Münstermann et al.’s (2010a, p. 31) definition of business process standardization as the “unification of variants of a given business process by aligning the variants against an archetype process”.

In the recruiting process, this means that standardization contributes to better recruiting process performance as HR employees can perform or operate a standardized recruiting process easier than a non-standardized one due to reduced overall process complexity and uncertainty (Lillrank and Liukko 2004; Münstermann et al. 2010a). Hence, standardizing recruiting reduces the cycle-time of the recruiting process. Moreover, process standardization reduces recruiting process costs by increasing operational efficiency and performance (Münstermann et al. 2010a). For example, Wüllenweber et al. (2008) observe that “decreasing process errors, facilitating communication, or just profiting from expert knowledge” are determinants of reduced process costs while standardizing recruiting processes. In addition, another positive effect of process standardization on business process quality is assumed as process standardization leads to operational excellence and improved operational performance, which then directly leads to higher business process quality (Münstermann et al. 2010a) and stakeholder satisfaction. Furthermore, subsidiaries and parent companies can mutually benefit each other’s experience to improve the different steps of the recruiting process continuously. Therefore, we hypothesize that:

Hypothesis 1c: *The higher the degree of business process standardization, the better the performance of the recruiting process.*

3.2.4 The impact of business process governance on process performance

Business process governance as another aspect of designing efficient and effective business processes (Dumas et al. 2013) refers to the “direction, coordination, and control of individuals, groups, or companies that are at least to some extent autonomous” (Markus and Jacobson 2010, p. 201) and “is necessary for the success of business processes” (Markus and Jacobson 2010, p. 201). For example, in human resource management processes, multiple organizational functions are involved such that the success of the recruiting process requires considerable lateral coordination across units, however it cannot easily be disentangled from the management of the people, functions, and companies that perform the activities making up the recruiting process (Markus and Jacobson 2010). If business process governance components are implemented and conducted to manage different people involved in the recruiting process, the cycle time of the process can be reduced as defined coordination reduces the uncertainty for recruiters of working with people from different operating departments (Braganza and Lambert 2000; Palmberg 2009). So we hypothesize that:

Hypothesis 1d: *The more transparent and structured the business process governance, the better the performance of the recruiting process.*

3.2.5 The impact of business process controlling on recruiting process performance

Besides the analysis of business processes, identifying performance measurements and targets to control the process is a key activity of BPM on the operational level (Dumas et al. 2013). This allows process improvements to be identified and implemented continuously (Palmberg 2009). *Business process controlling* is an ongoing activity that relies on standardized and defined measures to “continually monitor performance, assess results, and look for improvement opportunities” (Lee 1998, p. 218). In other words, business process analysis is a BPM component used to analyze the quantitative and monetary effects of the process to derive measures for process improvement.

We expect that companies controlling their recruiting process accurately and continuously using performance measures and key performance indicators have higher recruiting process performance. Such key performance indicators include “talent-on-time”, “brand strength”, “offer acceptance” and “quality of new hires” (Eckhardt et al. 2012). In the candidate attraction phase, the number of the job ad page visits and the number of online applications filled out completely by applicants can be monitored. This helps the company to calculate the average number of views needed per filled-out application for each media channel. Moreover, indicators can be established to control for what media channels candidates are attracted to and how many applications are submitted (Lee 2007; Holm 2012).

In business process controlling, it is important to track the time of each process step in order to identify process inefficiencies. Therefore, the time-to-hire can be split into granular single process steps to create transparency in terms of overall and individual process step time. Cost transparency is another important issue. Companies might gather information on the cost of open positions not filled as well as on the cost of each individual process step. To further improve the recruiting process, companies might also evaluate the candidate experience at individual process steps. During the recruiting process a companies might also ask candidates for reasons why they rejected a contract offer as a basis for making changes to increase future applicants’ general satisfaction with the recruiting process as well as to decrease process cost as fewer candidates turn down offers after or during the recruiting process. This avoids or reduces sunk costs for unsuccessful hires. For this reason, we hypothesize that:

Hypothesis 1e: *The higher the degree of business process controlling, the better the performance of the recruiting process.*

3.2.6 The impact of business process improvement on process performance

The final dimension of BPM, *business process improvement*, refers to a “structured approach to performance improvement that centers on the disciplined design and careful execution of a company’s end to-end business process” [(Hammer 2002), as cited by (Siha and Saad 2008, p. 779)] and also impacts business process performance. Business process improvement focuses on measures like six sigma (Siha and Saad 2008) that enable a continuously improvement of a business process

(Rohleder and Silver 1997; Niehaves et al. 2012). If a company is able to implement a structured approach for continuously improving its business processes the performance of its processes will increase (Adesola and Baines 2005).

If a company has implemented an environment and measures to frequently or, ideally, continuously optimize the recruiting process by focusing on process inefficiencies, the recruiting process performance can be improved in terms of reduced cycle time, reduced costs, increased quality and increased stakeholder satisfaction. Hence, we hypothesize that:

Hypothesis 1f: *The more frequent business process improvement measures are conducted, the better the performance of the recruiting process.*

3.3 Applicant tracking systems as a driver of recruiting process performance

While the previous section focuses on the components of BPM, this section deals with the related IT, such as an integrated applicant tracking system. An applicant tracking system (ATS) is an IS designed to support each task of the recruiting process (Lee 2007; Eckhardt et al. 2014). The concept of an ATS architecture for recruiting were first proposed by Palvia et al. (1992), while adopting a process-focused perspective in applicant tracking systems was proposed by Lee (2007). The ATS provides integrated IT support for the recruiting process by embedding all recruiting tasks into a single system. It supports the entire recruiting process from the identification of a vacancy to the on boarding of the hired candidate, providing a common data base for all process steps and facilitating the management of the recruiting process without any media disruption (Eckhardt et al. 2012). When a vacancy occurs, the hiring manager initiates the recruiting process using an interface provided by the applicant tracking system. The personnel requisition is forwarded to the HR department and submitted to the budget responsible for approval. After approval, the HR department and the hiring manager define the required skills and capabilities for the open position and design a job description, which is stored in a central database and can be used for further similar vacancies. Afterwards, the ATS creates a job advertisement which can be published on the corporate career website as well as on print and electronic job boards (Grund 2006; Froese and Peltokorpi 2011). Hiring and HR managers are able to determine the promotion-mix for the job advertisement using a central repository of different job boards conditions (Baum and Kabst 2012). Furthermore, the system should support the direct search for potential candidates in applicable databases or talent pools. In the third step of the recruiting process, an ATS should offer several interfaces for receiving and managing incoming applications (Laumer et al. 2010a). Moreover, the system can provide functionality that support the pre-selection and selection steps (Laumer et al. 2009a, b, 2012). The system facilitates cooperation between the hiring and the HR manager in order to select the most appropriate candidate for the open position. An ideal architecture of an ATS supporting the recruiting process is illustrated by Fig. 2.

The model ideal architecture of an ATS indicates that implementing IT to support the recruiting process might help streamline and structure the process to improve

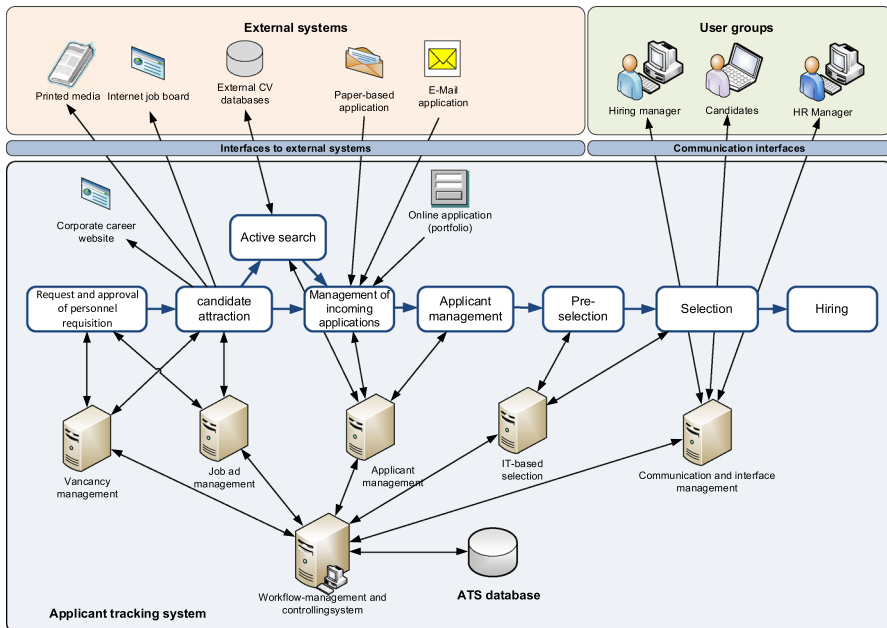


Fig. 2 Applicant tracking system and recruiting process (Eckhardt et al. 2012)

recruiting process performance. For example, when the workflow is managed electronically and the responsible managers are automatically reminded to complete tasks, the cycle time of the process can be improved as tasks are completed quicker and idle time can be reduced. Although the implementation of an ATS requires financial investment, it can also reduce process costs (Eckhardt et al. 2012). For example, negotiating a central contract with a global internet job board can reduce technical complexity (e.g. only one interface to an external service provider) and lower the cost of managing systems and processes. Furthermore, using standardized online application portfolios (e.g. Strohmeier 2010) can reduce the costs of managing the selection steps of the recruiting process. Unlike email and paper-based applications, standardized application portfolios can be processed without any media disruption, allowing fewer employees to handle more applications more easily and automating the pre-selection process (e.g. Laumer et al. 2009a, b). Hence, a company might save personnel costs by implementing an ATS as well. In terms of quality, an IT-based approach for managing and storing all information related to the recruiting process also increases the overall quality of the data used in the process. If the ATS provides strong support for the different tasks of hiring managers, HR managers, and candidates, their satisfaction with the process could increase. Hence, we hypothesize that:

Hypothesis 2: *Using an applicant tracking system has a positive impact on the performance of the recruiting process.*

3.4 The joint impact of business process management and information technology on recruiting process performance

Based on the proposed hypotheses, our model so far considers the independent influence of BPM and IT on recruiting process performance. However, there is also a combined impact of both concepts implemented in a company stemming from complementarities between the implemented components of BPM and the underlying ATS (Münstermann et al. 2009; Beimborn and Joachim 2011).

For example, process standardization as part of BPM might be a necessary precondition for implementing integrated applicant tracking systems to support a recruiting process free of media disruptions. An ATS relies on the documentation and standardization of the recruiting process to support the workflow management of the different defined process steps and hence internal process standards (Lee 2007). At the same time, the company might also need to implement external process standards in order to fully leverage the applicant tracking systems (e.g. process flow implemented in the ATS). Hence, business process standardization is an example of a BPM component that might be induced by technology implementation but might also be a precondition for implementing an ATS successfully. Hence, the full impact of BPM and IT on recruiting process performance might only be realized in combination. Hence, we further propose a third hypothesis:

Hypothesis 3: *The joint use of BPM and IT has a combined positive impact on the performance of the recruiting process.*

The research model based on the three proposed hypotheses is illustrated by Figs. 3, 4. In the section below, the hypotheses will be evaluated by a representative study of Germany's 1,000 largest companies.

4 Empirical evidence

We have evaluated our research model as proposed in the previous section with a representative empirical study focusing on the Germany's top 1,000 companies. This section describes the research methodology and the survey instrument used, and presents the results of a model validated using linear regression analysis.

4.1 Research methodology

To validate the research model, we conducted an empirical study in 2012 of Germany's top 1,000 companies by reported revenue, based on Hoppenstedt's database. The identified companies were contacted by telephone in order to identify the responsible manager of the recruiting process. If the recruiting manager agreed to participate in our study, a survey was sent to the company either by e-mail or by postal mail. A link to the survey was also provided so that the recruiting manager could participate in the survey via a website. The survey responses were entered into the statistic software SPSS and the research model was validated through linear

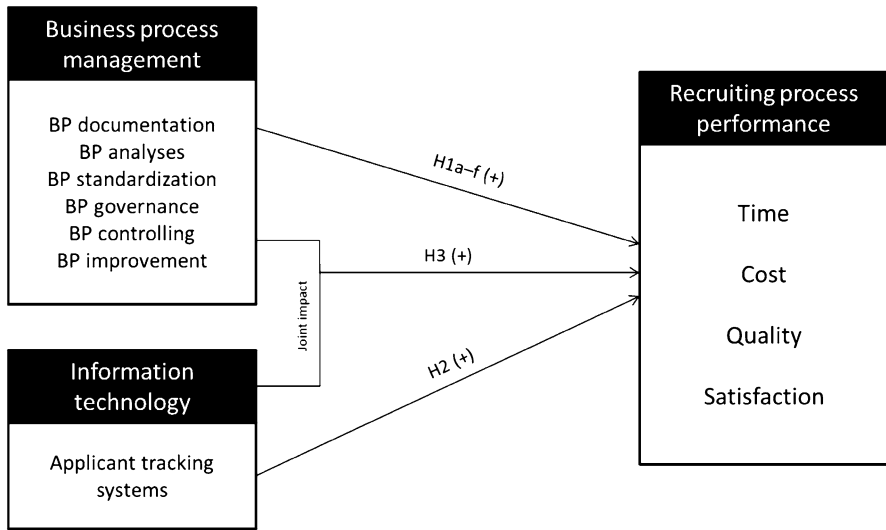


Fig. 3 Research model

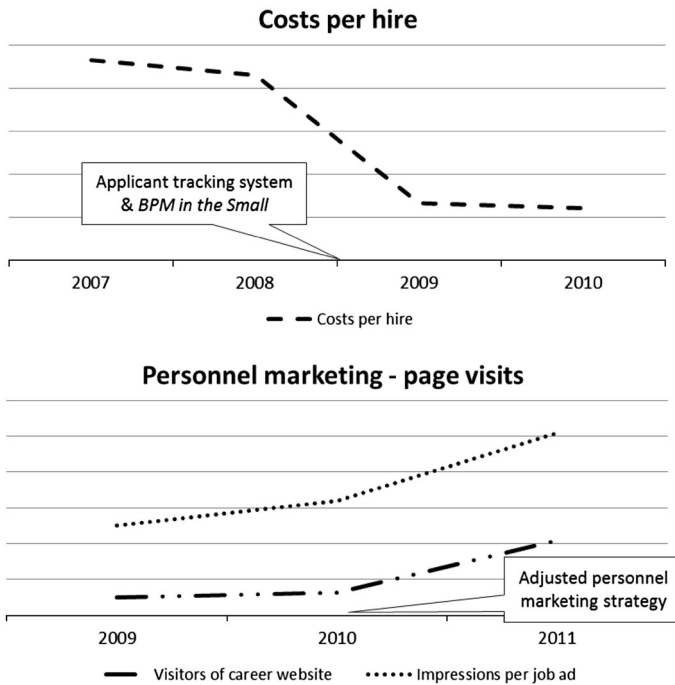


Fig. 4 Benefits of BPM and ATS implementation (Laumer et al. 2014)

regression analysis using SPSS 20. In total, 131 companies participated in our study (response rate 13.1 %). Table 3 illustrates the reported revenue, number of employees and the industry sector of the participating companies. We conducted a test of whether the 131 companies participating were statistically representative of Germany's top 1,000 companies. The significance level of the conducted Chi-quadrat-test (Mantel 1963) was above the recommended level of 0.050 for the three main criteria revenue ($p = 0.458$), number of employees ($p = 0.469$), and industry sector ($p = 0.237$). Consequently, the null hypothesis that the distribution of our sample statistically represents the population of Germany's top 1,000 companies cannot be rejected (Mantel 1963). Hence, the results based on the returned surveys of the 131 participating companies can be considered representative of the results for Germany's top 1,000 companies.

4.2 Measures

Based on prior BPM (Beimborn and Joachim 2011) and recruiting process performance studies (Münstermann et al. 2010a, b), we developed a survey instrument to capture the degree of BPM and IT usage as well as statements about the respective recruiting process performance.

Based on Beimborn and Joachim (2011), we used two items for business process documentation focusing on the degree of documenting the process steps of the recruiting process as well as the availability of these documents in a central repository. For business process standardization, two items were used based on Wüllenweber et al. (2008) and Münstermann et al. (2010a) capturing the degree of establishing highly standardized working routines in the recruiting process. Moreover, business process analysis was measured using two items based on Beimborn and Joachim (2011). These items focus on the degree of quantitatively and monetarily analyzing the recruiting process. In addition, two items were designed based on the quality control items of Beimborn and Joachim (2011) for business process controlling. These items focus on the definition of key performance indicators and the corresponding methods to measure these indicators. Business process governance was measured according the underlying work by Markus and Jacobson (2010). The items were transferred to the recruiting context to fit in our approach. The items for business process improvement were adapted from the work of Adesola et al. (2005) and Siha et al. (2008). To measure the use and ownership of an integrated ATS we took the items by Eckhardt et al. (2012). Finally, recruiting process performance in terms of time, cost, quality, and stakeholder satisfaction was measured according Münstermann et al. (2010a) as well as Hirzel (2008). All measurement items were captured using a 7-point Likert scale ranging from "strongly disagree" to "strongly agree".

4.3 Pre-study and content validity

In setting up the questionnaire, the aim was to refer to methods of measurement which had already been used in empirical research, as described above. The items were tested in a two-step process. After a first test with students in the information

Table 3 Characteristics of sample companies and 1,000 largest German companies

Revenue	Top 1,000 (in %)	Sample (in %)	Employees	Top 1,000 (in %)	Sample (in %)	Industry sector	Top 1,000 (in %)	Sample (in %)
Up to 500	4.9	4.6	Up to 100	6.5	6.8	Agriculture and forestry	0.3	0.8
501 to 600	12.7	9.2	101 to 400	9.2	8.3	Manufacturing	36.6	41.2
601 to 700	10.4	9.9	401 to 800	8.2	6.8	Energy	7.3	8.4
701 to 800	8.7	7.6	801 to 1,500	10.3	7.6	Building	1.4	1.5
801 to 950	9.1	7.6	1,501 to 2,300	10.6	8.3	Commerce	24.7	15.3
601 to 1,300	12.7	17.6	2,301 to 3,500	12.7	10.6	Hospitality	0.6	1.5
1,301 to 1,800	10.6	6.9	2,501 to 5,000	12.2	15.9	Communication	6.7	6.9
1,801 to 2,500	7.8	9.2	5,001 to 8,000	10.4	11.4	Finance	10.4	11.5
2,501 to 4,000	8.2	12.2	8,001 to 20,000	10.8	16.7	Service	9.6	8.4
4,001 to 10,000	5.2	6.9	More than 20,000	9.0	7.6	Public service	2.3	4.6
More than 10,000	9.8	8.4	Chi-quadrat-test	0.469		Chi-quadrat-test	0.237	
Chi-quadrat-test	0.458							

systems department of our university, the questionnaire was refined based on interviews with HR professionals in one company to ensure content validity. For this purpose, a company was chosen that has implemented both different components of BPM and an ATS (see Laumer et al. 2014 for an extended description of the case study). The company has followed a BPM approach including business process documentation, standardization, analysis, controlling, governance, and improvement, and adopted an ATS in 2009. This ensured that the responsible managers were able to discuss different BPM components as well as their impact on recruiting process performance in order to ensure the validity of our measurement model (Laumer et al. 2014).

In general, the implementation of the ATS as well as the different components of BPM have led to several improvements and value contributions for this company. The company was able to reduce their total cost in staff recruitment by fifty percent between 2007 and 2010. Especially the cost for job postings and the cost per hire could be reduced, as depicted in Fig. 4 (top) (Laumer et al. 2014). Furthermore, the joint BPM and ATS helped the company realize value potential in personnel marketing. The number of visitors to the career website and the impressions per job ad increased significantly (see Fig. 4 bottom). Based on the key performance indicators set in the process controlling, the company found out that adjustments in the personnel marketing strategy (e.g., reduced number of offline job ads and increased use of online job boards) led to more visitors on the career website and more impressions per job ad (Laumer et al. 2014).

We discussed our questionnaire and our research model with the responsible managers in this company to ensure content validity of the measurement items used. After the responsible manager provided a general description of their own project, we described our hypotheses and the corresponding measurement items. Based on his comments, we adjusted the wording of some items to better meet the BPM notation in the recruiting context. These adjusted measurement items are listed in Table 4 above and were used in our study with Germany's top 1,000 companies as described above.

4.4 Research results

This section provides statistics illustrating the reliability of our measurement instrument used and the results of our hypothesis testing.

4.4.1 *Measurement instrument reliability and validity*

To ensure the reliability and validity of the measurement instrument issued, we conducted several tests. First, indicator reliability is confirmed as the factor loading tests reveals that all but two items are above the recommend threshold of 0.7 (Hulland 1999), and these two items are still higher than the 0.6 threshold recommended by Bagozzi and Yi (1988). Further, construct reliability was operationalized by calculating Cronbach's Alpha, which exceed the value of 0.7 recommended by Hair et al. (2010) indicating sufficient construct reliability (see Table 5).

Table 4 Measurement model

Dimension	Items
BP documentation (Beimborn and Joachim 2011)	We have documented almost all process steps of our recruiting process A documentation of our recruiting process is available in a central repository
BPstandardization (Wüllenweber et al. 2008, 2010a, b)	We have established highly standardized procedures in our recruiting process We have realized a high degree of standardization of all steps in our recruiting process
BP analysis (Beimborn and Joachim 2011)	Our recruiting process has been quantitatively analyzed (process time) Our recruiting process has been monetarily analyzed (process costs)
BP controlling (Beimbornand Joachim 2011)	We have defined key performance indicators of our recruiting process We have defined methods to measure the key performance indicators of our recruiting process
BP governance (Markus and Jacobson 2010)	We have appointed a process owner of the recruiting process We have established clear responsibilities of all process stakeholders
BP improvement (Adesolaand Baines 2005; Siha and Saad 2008)	We have implemented measures to continuously improve our recruiting process We have established routines to continuously improve our recruiting process
Applicant tracking system (ATS) (Eckhardt et al. 2012)	Our recruiting process is completely supported by a single enterprise system (e.g. applicant tracking system) All process steps of our recruiting process are integrated by a single applicant tracking system
Cost (Münstermann et al. 2010a, b)	We have been able to reduce the total costs of the recruiting process in recent years We have been able to reduce the costs for the internal applicant tracking in recent years We have been able to reduce the costs per tracked application in recent years
Time (Münstermann et al. 2010a, b)	We have reduced the time between identifying a vacancy and filling it in recent years We have reduced the time between posting a job ad and making the final hiring decision in recent years We have reduced the time between identifying a vacancy and holding the first job interviews in recent years
Quality (Münstermann et al. 2010a, b)	We have been able to enhance the overall applicant quality in recent years We have been able to enhance the quality of applicant data in recent years We have been able to enhance the overall quality of the recruiting process in recent years

Table 4 continued

Dimension	Items
Satisfaction (Hirzel 2008)	The operating departments are very satisfied with our recruiting process Our HR managers are very satisfied with our recruiting process Our applicants are very satisfied with our recruiting process Our managing directors are very satisfied with our recruiting process

7-point Likert scale ranging from “strongly disagree” to “strongly agree” were used (1 = strongly disagree, 7 = strongly agree); “We” and “our” refers to the recruiting department of the respective company; “Recent years” refers to a time period of 5 years as indicated in the introductory sentence in the questionnaire

To ensure convergent and discriminant validity, a confirmatory factor analysis was conducted. This analysis shows that for the seven independent and the four dependent variables each item was assigned to its respective variable.

To reduce the problem of multicollinearity between the predictor and the impendent variables, we standardized the predictor variables (Cohen 1988; Chin et al. 2003) and calculated that the VIF-values were below the threshold of 3.33 (Diamantopoulos and Siguaw 2006). We also tested the potential impact of common method variance using the Harman Single Factor test, which resulted in a first component variance of 0.36, such that no sign of CMB could be observed (Podsakoff et al. 2003). Hence, these tests indicates a good reliability and validity of the measurement model used.

In addition to the measurement model validation Table 6 provides an overview of the mean, standard derivation, and the correlation between the BPM components, the usage of an ATS and the dimensions of recruiting process performance.

4.4.2 Hypotheses testing

In order to test for the proposed influence of BPM, IT, and the joint impact of both concepts, we performed three linear regression analyses to test our hypotheses in three steps. In the first step, the variables modeling the degree of BPM were integrated into the model. Next, the variable reflecting the usage of applicant tracking systems was integrated into each separate model. Finally, the interaction terms were entered to each regression analysis. Table 7 below summarizes the results of the three linear regression analyses for each performance dimension, which are discussed in the following, italicized the results of the effects that are significant as recommended by Kohlbacher (2010).

For the first dimension, the results of the first linear regression model based on the representative study of 131 German companies illustrate that business process analysis and business process improvement have a significant positive impact on reduced costs of the recruiting process. Moreover, the second linear regression model illustrates that the usage of applicant tracking systems also has a positive significant impact on the cost of the recruiting process. These results are confirmed

Table 5 Measurement model reliability

	Loadings	Cronbach's alpha
<i>Independent variables</i>		
BP documentation	0.891	0.84
	0.753	
BP standardization	0.638	0.91
	0.724	
BP governance	0.902	0.92
	0.891	
BP controlling	0.925	0.96
	0.917	
BP improvement	0.732	0.85
	0.844	
BP analysis	0.627	0.85
	0.708	
ATS	0.877	0.94
	0.949	
<i>Dependent variables</i>		
Cost	0.752	0.87
	0.902	
	0.851	
Time	0.854	0.90
	0.831	
	0.841	
Quality	0.645	0.72
	0.880	
	0.771	
Satisfaction	0.771	0.90
	0.886	
	0.837	
	0.851	

by the third linear regression model including the interaction terms. For process costs, no significant effect of an interaction of applicant tracking systems and BPM can be observed.

For the quality dimension, the results of the linear regression model show that business process controlling has a positive significant impact on the quality of the recruiting process. This effect is also confirmed by the second and third linear regression model. Although using applicant tracking systems does not have a direct effect on the quality of the recruiting process, the third linear regression model reveals a significant positive interaction effect of using applicant tracking systems and business process controlling. In other words, even though applicant tracking systems alone do not improve the recruiting process, using applicant tracking systems and business process controlling in combination leads to higher

Table 6 Descriptive statistics and correlation analysis

	Mean	Std. dev.	1	2	3	4	5	6	7	8	9	10	11
1 BP documentation	4.27	1.61	1										
2 BP standardisation	4.90	1.44	0.692**	1									
3 BP analysis	3.39	1.55	0.469**	0.576**	1								
4 BP controlling	3.38	1.62	0.421**	0.463**	0.788**	1							
5 BP governance	4.06	1.62	0.322**	0.602**	0.430**	0.387**	1						
6 BP improvement	4.98	1.41	0.523**	0.492**	0.605**	0.567**	0.503**	1					
7 Applicant tracking system	3.94	2.42	0.475**	0.463**	0.359**	0.391**	0.389**	0.282**	1				
8 Cost	4.42	1.46	0.315**	0.291**	0.405**	0.322**	0.149 ^{ns}	0.375**	0.339**	1			
9 Quality	4.75	1.10	0.285**	0.279**	0.396**	0.426**	0.182 ^{ns}	0.339**	0.203*	0.525**	1		
10 Satisfaction	5.11	1.06	0.226*	0.391**	0.346**	0.302**	0.363**	0.335**	0.217*	0.374**	0.435**	1	
11 Time	4.60	1.40	0.140 ^{ns}	0.250*	0.468**	0.451**	0.244*	0.291**	0.147	0.475**	0.574**	0.367**	1

** $p < 0.01$, * $p < 0.1$, ns $p > 0.1$

Table 7 Research results (linear regression analyses, $N = 131$; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$, ^{ns} $p > 0.1$)

Research results	Reduced cost			Increased quality			Increased satisfaction			Reduced time		
	Beta	R ²	ΔR^2	Beta	R ²	ΔR^2	Beta	R ²	ΔR^2	Beta	R ²	ΔR^2
BPM												
BP documentation	0.125 ^{ns}	0.190		0.010 ^{ns}	0.201		-0.155 ^{ns}	0.225		-0.270 ^{ns}	0.252	
BP standardization	-0.030 ^{ns}			0.036 ^{ns}			0.421***			0.127 ^{ns}		
BP analysis	0.418**			0.055 ^{ns}			0.102 ^{ns}			0.339*		
BP controlling	-0.039 ^{ns}			0.302***			0.085 ^{ns}			0.378**		
BP governance	-0.071 ^{ns}			-0.068 ^{ns}			-0.044 ^{ns}			-0.111 ^{ns}		
BP improvement	0.248 ^{ns}			0.145 ^{ns}			0.082 ^{ns}			0.051 ^{ns}		
ATS												
BP documentation	0.044 ^{ns}	0.221	0.031***	0.004 ^{ns}	0.201	0.000	-0.156 ^{ns}	0.225	0.000	-0.256 ^{ns}	0.253	0.001
BP standardization	-0.078 ^{ns}			0.032 ^{ns}			0.420***			0.135 ^{ns}		
BP analysis	0.422**			0.055 ^{ns}			0.102 ^{ns}			0.338**		
BP controlling	-0.114 ^{ns}			0.296***			0.084 ^{ns}			0.391**		
BP governance	-0.099 ^{ns}			-0.070 ^{ns}			-0.044 ^{ns}			-0.106 ^{ns}		
BP improvement	0.291*			0.148 ^{ns}			0.083 ^{ns}			0.044 ^{ns}		
Applicant tracking system (ATS)	0.291***			0.024 ^{ns}			0.004 ^{ns}			-0.051 ^{ns}		

Table 7 continued

Research results	Reduced cost			Increased quality			Increased satisfaction			Reduced time		
	Beta	R ²	ΔR^2	Beta	R ²	ΔR^2	Beta	R ²	ΔR^2	Beta	R ²	ΔR^2
BPM × ATS												
BP documentation	-0.037 ^{ns}	0.254	0.033***	0.073 ^{ns}	0.252	0.051***	-0.110 ^{ns}	0.319	0.094***	-0.293 ^{ns}	0.301	0.048***
BP standardization	-0.006 ^{ns}			-0.130 ^{ns}			0.290*			0.088 ^{ns}		
BP analysis	0.449**			0.114 ^{ns}			0.180 ^{ns}			0.396**		
BP controlling	-0.111 ^{ns}			0.229 ^{ns}			-0.009 ^{ns}			0.367**		
BP governance	-0.119 ^{ns}			-0.062 ^{ns}			-0.060 ^{ns}			-0.108 ^{ns}		
BP improvement	0.318*			0.188 ^{ns}			0.126 ^{ns}			0.116 ^{ns}		
Applicant tracking system (ATS)	0.265*			0.057 ^{ns}			0.038 ^{ns}			-0.062 ^{ns}		
ATS × BP documentation	-0.058 ^{ns}			0.158 ^{ns}			0.233*			-0.123 ^{ns}		
ATS × BP standardization	0.178 ^{ns}			-0.153 ^{ns}			-0.120 ^{ns}			0.089 ^{ns}		
ATS × BP analysis	-0.059 ^{ns}			-0.330 ^{ns}			-0.289 ^{ns}			-0.385 ^{ns}		
ATS × BP controlling	0.261 ^{ns}			0.284*			0.448***			0.445***		
ATS × BP governance	-0.168 ^{ns}			0.064 ^{ns}			0.024 ^{ns}			-0.106 ^{ns}		
ATS × BP improvement	0.019 ^{ns}			-0.079 ^{ns}			-0.179 ^{ns}			0.021 ^{ns}		

performance of the recruiting process in terms of quality. The results of this interaction effect are illustrated in Fig. 4 below.

For the third dimension, satisfaction, the results of the three linear regression models reveal a significant impact of business process standardization on the satisfaction of recruiting process stakeholders. Although, no significant impact of applicant tracking systems on stakeholders satisfaction can be observed (see second linear regression model), the results of the third linear regression model reveals two significant interaction effects. The first illustrates that a joint usage of applicant tracking systems and business process documentation has a significant impact on stakeholder satisfaction. The second reveals that using applicant tracking systems enables a significant impact of business process controlling on the performance of the recruiting process in terms of stakeholder satisfaction. These interactions effects are illustrated in Fig. 5 below.

For the time dimension, the results of the linear regression models reveal a positive impact of business process analysis and business process controlling. Moreover, the third model indicates a positive interaction effect of applicant tracking systems and business process controlling, even though the second linear regression model shows no significant effect of using an ATS alone. Hence, using

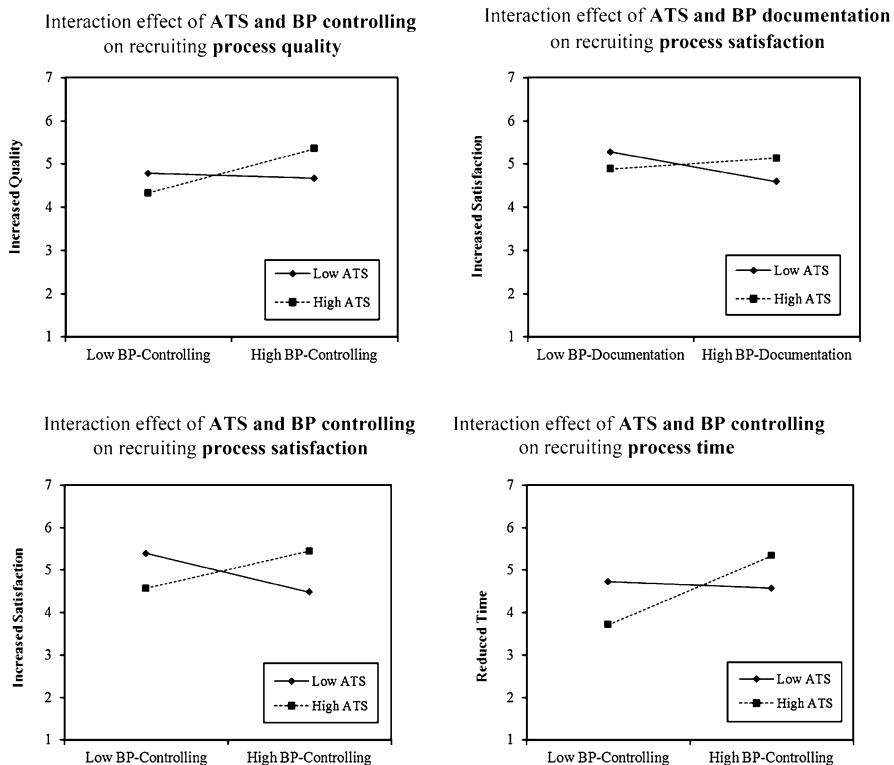


Fig. 5 Joint impact of business process management and applicant tracking systems (1 = strongly disagree, 7 = strongly agree)

applicant tracking systems enables a significantly greater impact of business process controlling on the performance of the recruiting process in terms of time. The observed interaction effect is illustrated in Fig. 5 below.

In summary, the results of the linear regression analyses based on the representative study of 131 German companies reveal that BPM and the usage of applicant tracking systems have a positive impact on recruiting process performance, and the joint effect of implementing both BPM and applicant tracking systems enables a significant better performance of the recruiting process in the companies observed.

The results of the empirical study will be discussed in the concluding section.

5 Discussion, implications, and future research

“No one seems to have developed a business process model of the HRM business processes” (Cakar et al. 2003, p. 191), although “treating the HRM process as a support process providing critical support to the key valued-add processes” (Cakar et al. 2003, p. 191). With this survey, we have taken a “BPM in the small” approach to evaluate the independent and combined impacts of BPM and IT on recruiting process performance.

The results of our representative study of Germany’s top 1,000 companies ($N = 131$) reveal that a reduction of recruiting process costs is significantly determined by business process analysis, business process improvement, and the usage of applicant tracking systems. Reduced cycle time of the recruiting process is significantly influenced by business process analysis, business process controlling and by the usage of an ATS that supports the controlling tasks. Business process controlling and a joint usage of applicant tracking systems and business process controlling has a significant positive impact on the quality of the recruiting process. The satisfaction of the respective stakeholders of the process can be improved through business process standardization and a joint usage of applicant tracking systems, business process controlling and business process documentation. In summary, the results reveal that different components of the BPM in conjunction with IT have diverse impacts on different dimensions of recruiting process performance. These results, summarized in Table 8, will be discussed in more detail in the following, focusing on the diverse effects of BPM on process performance and the joint impact of BPM and applicant tracking systems on recruiting process performance.

5.1 Diverse effects of business process management

This study is a response to the identified need of studies exploring which individual BPM components effect business process performance to what degree (Kohlbacher 2010). Whereas in several case studies the positive effect of BPM is reported (e.g. Armistead and Machin 1998; Hertz et al. 2001; Küng and Hagen 2007; Münstermann et al. 2010b), quantitative studies focusing different BPM components are seldom. Therefore, the current study extends the BPM literature by

Table 8 Summary of results

	Recruiting process performance			
	Cost	Time	Quality	Satisfaction
H1a: BP documentation				
H1b: BP analysis	Supported	Supported		
H1c: BP standardization				Supported
H1d: BP governance				
H1e: BP controlling		Supported	Supported	
H1f: BP improvement	Supported			
H2: ATS	Supported			
H3: ATS $\times$ BP documentation				Supported
H3: ATS $\times$ BP controlling		Supported	Supported	Supported

providing an empirical study of the positive effects on process performance and by highlighting the diverse effects BPM might have. Prior research has not always explicitly differentiated between the deployment of various BPM components and their effects on process performance. Based on our study, we can conclude that the impact on different performance dimensions in the recruiting process varies depending on the components of BPM, as outlined above.

Recognizing the limitation that our results are only derived from one empirical study focusing on one secondary business process, the confirmed effects for this scenario indicate that business process analysis and improvement have an effect on process costs, controlling on process time, standardization on process quality, and controlling and analysis on stakeholders satisfaction. Based on this derived extended understanding of BPM components, we can take a first step toward filling the identified gap in research providing empirical confirmed explicit differentiation of the effect of various BPM components on process performance (Kohlbacher 2010). Our study indicates the positive effect of business process analysis, business process improvement and the implementation of IT on process cost, of business process analysis and IT-supported business process controlling on process time, of an IT-supported controlling on process quality, and of business process standardization and an joint implementation of IT and business process documentation as well as controlling on business process stakeholders satisfaction.

Nevertheless, we also observed non-significant effects of several BPM components. All tested linear regression models indicate that business process documentation and business process governance have no significant direct impact on business process performance. Based on our study, we cannot conclude that those components directly effect process performance. Nonetheless, these results are in line with the call by Kohlbacher (2010) as they illustrate the various effects BPM components might have on process performance. One reason for the observed effects might be that these dimensions have positive effects on different outcome variables not tested in our model. In addition, there may have been indirect effects as both components might impact others, which in turn influence process

performance. Finally, the effects might be different for other processes and organizational settings.

Comparing our results with other research studies focusing on other business processes we have to acknowledge that our results decrease the importance of business process documentation as no significant effect can be observed. Other studies reveal a high importance for this BPM component especially in primary business process (e.g. Ungan 2006; Beimborn et al. 2008). Nonetheless, we observe a significant correlation of business process documentation with other BPM components which have an impact on process performance. For example, documentation is strongly related with standardization and improvement which have both an impact on performance. This is in line with the results of Ungan (2006) who indicates that business process standardization can be realized through business process documentation or Beimborn et al. (2008) who reveal documentation as an important aspect of IT/Business-Alignment. Hence, business process documentation is one example responding to the critique of Kohlbacher (2010) who states that most studies do not state the exact BPM efforts implemented such that our differentiation highlights that the positive effects of the recruiting process are mainly determined by the joint implementation of BPM in terms of and IT, nonetheless, other non-significant components are strongly related to these components and a better understanding of the diverse components is necessary.

As our model was not able to control for all variables and it has not been tested in other processes than the recruiting one, we would like to encourage future research to extend our study by focusing on different organizational settings and different scenarios in order to further extend the understanding of the diverse effects of BPM on process performance. We assume that the purpose of the analyzed process will moderate the impact of various BPM components on process performance such that the effects will differ for core and secondary processes. Furthermore, the interplay between the different BPM components need further theorizing and empirical evidence. In addition, as we are limited by self-reported data only we encourage future research to strengthen our argumentation by focusing on longitudinal and objective performance data. Nonetheless, our results already indicate that the effects of different BPM components on process performance are diverse which addresses the shortcomings of prior BPM research in a first step as it highlights especially the joint effect of BPM and IT implementations.

5.2 Joint effect of business process management and information technology

We can conclude from our study that the impact implementing both BPM and IT is stronger than the impact of implementing either BPM or IT. As Fig. 5 illustrates, implementing an ATS in conjunction with business process controlling or business process documentation has a stronger effect on the satisfaction dimension of the recruiting process. Based on the fact that implementing only an ATS does statistically not impact the performance of the recruiting process, the results illustrate that only joint and integrated implementation of BPM and IT enables significant positive effects for the quality of the recruiting process.

This result indicates that companies that aim to improve stakeholder satisfaction cannot realize any significant positive effect by either implementing BPM in terms of process controlling or documentation or by adopting an applicant tracking system. In order to gain positive effects by implementing an applicant tracking system, companies have to integrate BPM and IT. These results are echoed by the joint effect of implementing an ATS and business process controlling on improved process quality and time. The sole effect of business process controlling is strengthened by the implementation of an applicant tracking system. Thus, companies implementing an applicant tracing system in conjunction with business process controlling benefit more than companies only implementing business process controlling. We conclude that IT significantly supports the effect of business process controlling on process performance. Based on these results, we can derive the recommendation that when implementing an applicant tracing systems as part of a transformation of the recruiting process (Eckhardt et al. 2014) companies should implement business process management for the recruiting process in parallel. In addition, as especially the combination of an ATS and business process controlling has positive effects on process time, quality, and stakeholder satisfaction, software vendors developing and distributing applicant tracking systems should focus on controlling functionality as a lever for achieving positive recruiting process performance effects.

The implication of our study for IS research is that an important variable is missing from the rich tradition of research of IS success (DeLone and McLean 1992; Petter et al. 2008; Dwivedi et al. 2014) that would better explain how a newly implemented information system can be more successful than the already confirmed effects. The most prominent model—the IS Success Model (DeLone and McLean 2003)—highlights that IS success is determined by information quality, system quality, and service quality of the information system. Based on our results, we control for these technology-focused impacts with the variable of using ATS in the recruiting context revealing a positive impact on process costs. Beyond that, we observed an even stronger effect of a combined implementation of BPM and IT on process performance. Hence, the success of the information systems is strengthened by the implementation of BPM components. Based on our study, we can conclude that companies implementing a new information system (here: applicant tracking system) benefit from system implementation by implementing and integrating BPM in parallel. Hence, we conclude that success is determined by both the IT and the combined implementation of BPM which constitutes an extension of previous IS success models lacking the BPM component.

To further extend the IS success model, we suggest that future research further theorize a “business process management quality” variable that focuses on the BPM activities implemented parallel to an information system. Recognizing the specific context of our single-study research focusing on a single process, we can only confirm this effect for the recruiting context in Germany’s top 1,000 companies and call for research building upon this idea to evaluate the joint effect of BPM and IT on IS success in further studies focusing on different process and different enterprise systems, such as enterprise resource planning (Karimi et al. 2007), social software (Maier et al. 2014), or enterprise content management (Laumer et al. 2013).

Furthermore, as the interaction observed in our study does not test whether the implementation of IT is moderated by BPM concepts or the other way round, we only can observe a general interaction effect. Hence, further studies can build upon our results by developing a more detailed theoretical explanation of the interaction between IT and BPM.

5.3 Conclusion

Kohlbacher (2010) called for investigation into which BPM components have an effect on which outcome variables of a business process, noting that prior research has not always explicitly differentiated between the deployment of various BPM components and their effects on process or organizational performance. In case of the recruiting process and based on our study of Germany's top 1,000 companies ($N = 131$), we are able to conclude that different BPM components have diverse effects on different business process performance variables. Our study illustrates the joint impact of BPM and IT on the success of an IS.

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